Manu Nicholas Jacob

Hardware Engineer, Root Cause Engineering · Dell Technologies · Austin, Texas

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Hardware engineer on enterprise AI server platforms: system-level and board-level debug, failure analysis and root-cause engineering across PCIe, memory, power and GPU subsystems. Built a server benchmarking and power-telemetry platform that cut a 4-hour manual process to 100 minutes. Named inventor on an invention disclosure authorized for filing, and an independent researcher with 14 manuscripts, 13 under peer review at IEEE and ACM venues.

WORK EXPERIENCE

Hardware Engineer, Root Cause Engineering · **Dell Technologies, Austin, TX** — Jun 2025 to present

- Root-caused system-level and board-level failures on enterprise server platforms spanning memory, PCIe, power delivery, CPU and management-controller subsystems, isolating faults by controlled reproduction rather than component swapping.
- Diagnosed a riser detection failure down to a single open boost-converter inductor using an oscilloscope, logic analyzer and micro X-ray imaging, proving mechanical solder-joint separation rather than electrical overstress.
- Separated genuine component failures from system-level interactions across recurring controller errors and repeat no-boot cases, closing several as no-defect-found on reproducible evidence.
- **On a performance-analysis rotation within the role** (Feb to Apr 2026), built an automated server benchmarking and power-telemetry platform that replaced a 4-hour manual workflow with a 100-minute run, **62% faster** and 5 minutes of operator time.
- Orchestrated a 10-phase load sequence for that platform, sampling OS metrics every 2 seconds and platform power every 8 across 90+ sensors, with automatic reconnection and dual-write persistence to SQLite and PostgreSQL.

Root Cause Engineering Intern · **Dell Technologies, Austin, TX** — Summer 2024

- Developed a unified diagnostic tool for NVIDIA AI platforms that detected PCIe replays, bus resets and GPU faults through a single workflow.
- Implemented optimized secondary-bus-reset procedures and real-time PCIe replay detection, improving fault isolation and technician usability across platform debug.
- Partnered with platform engineers and subject-matter experts to cut link retrain time and rework the tool's interface for technician use on the debug bench.

Controls Engineering Intern · **Schneider Electric, Lexington, KY** — Summer 2023

- Designed and deployed a human-machine interface for palletizer control on the factory floor, converting complex PLC output into an operator-facing interface for real-time monitoring and decision-making.
- Optimized PLC-to-HMI communication on the packer line, improving line efficiency and reducing downtime through more reliable data exchange.
- Led an overhaul of internal standardization specifications against ANSI/RIA and OSHA requirements, adding a dedicated section for robotic cells.

ENGINEERING PROJECTS

- **Pothole Detection System** · ECE senior design capstone, UMass Amherst, Sep 2024 to May 2025. Achieved 80 to 90% detection accuracy at 35 to 45 mph by fusing stereo depth and ultrasonic sensing into CNN-based detection on Raspberry Pi, with a custom PCB handling power, sensor interfacing and Pi connectivity, and an in-vehicle display giving the driver an immediate alert.
- **WorldWide Rover** · Winner, Best Embedded System, HackUMass XII. Built an internet-controlled rover drivable from anywhere, with onboard obstacle avoidance and maze solving when unattended.
- **4Sight** · Winner, Best Hardware Hack and Best Circuit Hack, HackUMass X. Designed a wearable converting ultrasonic distance readings into haptic feedback, so obstacles are felt rather than heard.

PATENTS AND INVENTION DISCLOSURES

- **Named inventor on a Dell Technologies invention disclosure authorized for filing.** Formal USPTO filing pending; no application number issued yet.

SKILLS

Hardware: enterprise servers, PCIe, GPU subsystems, DDR memory, power delivery, board-level debug, oscilloscope, logic analyzer, X-ray inspection, embedded systems, microcontrollers, custom PCB.

Software: Python, C, C++, SQL, PostgreSQL, SQLite, Bash, Linux, ONNX Runtime, PyTorch, llama.cpp, Linux `perf`, hardware performance counters, Redfish, Git.

Engineering: root-cause analysis, failure analysis, hardware validation, reliability engineering, benchmarking and instrumentation, design of experiments, thermal and power characterisation, diagnostics development, technical and patent writing.

Certifications: NVIDIA AI Infrastructure and Operations Fundamentals (2026), Oracle Cloud Infrastructure 2025 Certified Generative AI Professional, OCI 2025 Certified AI Foundations Associate, OCI 2025 Certified Foundations Associate.

EDUCATION

University of Massachusetts Amherst · **B.S. Electrical Engineering and B.S. Computer Engineering** 2021 to May 2025

Minor in Engineering Management; Minor in Economics.

RESEARCH AND PUBLICATIONS

- **14 manuscripts, 13 currently under peer review** at IEEE Transactions on Computers, ACM Transactions on Embedded Computing Systems, IEEE Internet of Things Journal, IEEE Computer Architecture Letters, IEEE Embedded Systems Letters and IEEE Design & Test.
- 5 preprints on TechRxiv, 4 citations. Every result ships with a DOI-archived artifact, and every headline number is measured on silicon. Full record: manunicholasjacob.com/research.
- Selected: **The Memory Wall at the Edge of Language** (IEEE Transactions on Computers, decode roofline $R^2 = 0.994$, revalidated on x86) and **The Memory Wall Governs Edge DNN Inference** (ACM Transactions on Embedded Computing Systems, 20.5% leave-one-model-out error across 9 CNNs and a ViT).

OPEN SOURCE CONTRIBUTIONS

- **NVIDIA garak** (LLM vulnerability scanner): merged 3 of 11 submitted pull requests, covering detector config-root handling, plugin-loader error restoration and a probe ASCII-bounds guard.
- **vLLM**: authored 4 open pull requests for tokenizer special-token handling, mixture-of-experts benchmark tuner robustness and troubleshooting documentation, each arising from a bug root-caused and filed as an issue first.
- Published **llama-roofline**, a command-line tool measuring where a local language model sits against the memory-bandwidth roofline on the reader's own machine, and **ML Systems Lab**, a reproducible cross-hardware benchmarking framework. Both installable from PyPI and archived on Zenodo with citable DOIs.
- Published **8 DOI-archived research artifacts** on Zenodo, MIT licensed, each carrying the harness and data needed to reproduce a result.

PROFESSIONAL SERVICE

- **Artifact Evaluation Committee, USENIX ATC 2026**, a CORE A systems venue. Evaluating submitted research artifacts for functionality and reproducibility; review window 22 September to 14 October 2026.
- **Reviewer, Journal of Open Source Software**. Completed the review of Optiland (optical design in Python), full 31-item checklist; currently assigned to three submissions.
- **Judge and Judge Advisor, VEX Robotics**, repeat judge across V5 competition events including mentoring first-time panels. **Judge, HackTX 2025** (UT Austin), 700+ participants.
- **Founded and hosted *Path to 99 Percentile***, a podcast interviewing professionals internationally; 5 guest episodes live on Spotify and Apple Podcasts.
- **Taught and volunteered** with 3 foundations in India: Lovedale (mathematics and science, grades 7 to 10, plus board-exam coaching), Diya and Yuvalok.